

MGT 924: Statistical Foundations

Problem Set 1

Group: Ashley Wen, Emily Wu, Skye Wang, Yiting Huang

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1 Visualize before building a prediction rule (20 points)

1.1 Sample Statistics (1.1)

For each of the four datasets (I, II, III, IV), the following sample statistics were computed: the mean of $\mathbf{x}$, the mean of $\mathbf{y}$, the variance of $\mathbf{x}$, the variance of $\mathbf{y}$, the correlation between $\mathbf{x}$ and $\mathbf{y}$, and the R^2 implied by these statistics alone (using the identity $R^2 = r_{xy}^2$ for simple linear regression).

Table 1: Sample statistics by dataset

Dataset	Mean x	Mean y	Var x	Var y	Corr(x, y)	R^2
I	9	7.5009	11	4.1273	0.8164	0.6665
II	9	7.5009	11	4.1276	0.8162	0.6662
III	9	7.5000	11	4.1226	0.8163	0.6663
IV	9	7.5009	11	4.1232	0.8165	0.6667

1.2 OLS Estimates (1.2)

For each dataset, OLS estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ were obtained from a regression of $\mathbf{y}$ on $\mathbf{x}$ using `lm()`.

Table 2: OLS estimates by dataset

Dataset	$\hat{\beta}_0$ (Intercept)	$\hat{\beta}_1$ (Slope)
I	3.0001	0.5001
II	3.0009	0.5000
III	3.0025	0.4997
IV	3.0017	0.4999

1.3 Trading Rule from Statistics Alone (1.3)

Based solely on the statistics computed in 1.1 and 1.2, how would you trade each of the four stocks?

Your answer:

From the statistics, the linear relationship between x and y is approximately the same across all four stocks:

$$\hat{y} = 0.5x + 3,$$

with a positive correlation of approximately 0.816 and an R^2 of approximately 0.666. Therefore, I would enter long positions in all four stocks when x is high and short positions when x is low.

1.4 Scatterplots with OLS Prediction Rule (1.4)

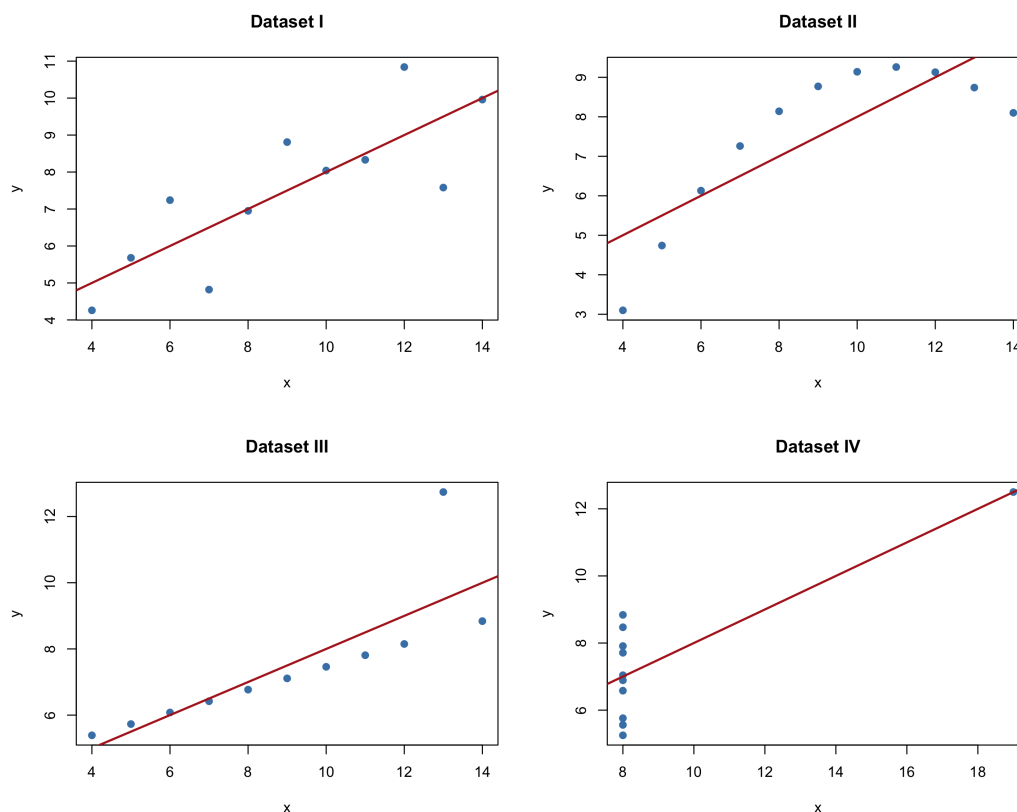


Figure 1: Scatterplots of y vs. x for each dataset, with the fitted OLS line overlaid.

1.5 Improving the Trading Strategy (1.5)

For each stock, describe the one improvement to your trading strategy (relative to your answer in 1.3) that the scatterplot suggests is most important.

Although the four stocks have nearly identical summary statistics and OLS estimates, their scatterplots reveal very different patterns in the relationship between x and y . Therefore, the common linear trading strategy proposed in 1.3 should be adjusted separately for each stock based on the structure revealed by the scatterplots.

Dataset I:

The data points are distributed roughly symmetrically around the fitted line, suggesting that the linear relationship provides a reasonable description of the data. Therefore, I would retain the original strategy from 1.3: go long when x is high and short when

x is low.

Dataset II:

The scatterplot shows a clear nonlinear, inverted-U-shaped relationship. y initially increases with x , but after reaching a peak, the relationship becomes negative.

Therefore, I would use different trading rules on the two sides of the turning point: go long as x rises before the peak, but reverse the strategy after the peak and short as x rises further.

Dataset III:

The scatterplot shows that one unusual observation pulls the fitted line upward and makes the positive relationship between x and y appear stronger than it is for most of the data. Therefore, I would lower my expectation that higher x reliably predicts higher returns. I would still go long when x is high and short when x is low, but with smaller positions and less confidence in the signal.

Dataset IV:

In this plot, for most observations, there is no clear relationship between x and y . The apparent positive linear relationship is driven almost entirely by a single high-leverage observation.

Therefore, I would not rely on the fitted linear rule. And I would avoid taking long or short positions based on x alone without additional evidence.

2 Predicting house prices (40 points)

2.1 Data Preparation (2.1)

The following code loads the Connecticut property data, constructs the true assessed value, expresses both assessed value and sale amount in \$1,000s, and filters the sample to single-family homes in New Haven listed in 2008 with true assessed values below \$250,000.

```
# ---- 2.1.1: Load the data ----
ct <- read.csv("ps1/ct_property_data.csv")

# ---- 2.1.2: Create assessedvalue_true ----
# Tax-assessed value in CT is 70% of true market value,
# so dividing by 0.7 recovers the true value.
ct$assessedvalue_true <- ct$assessedvalue / 0.7

# ---- 2.1.3: Express in $1,000s ----
ct$assessedvalue_true <- ct$assessedvalue_true / 1000
ct$saleamount <- ct$saleamount / 1000

# ---- 2.1.4: Filter ----
```

```
ct_filtered <- ct[
  ct$residentialtype == "Single Family" &
  ct$town == "New Haven" &
  ct$listyear == 2008 &
  ct$assessedvalue_true < 250,
]
```

The resulting sample contains **279 observations**.